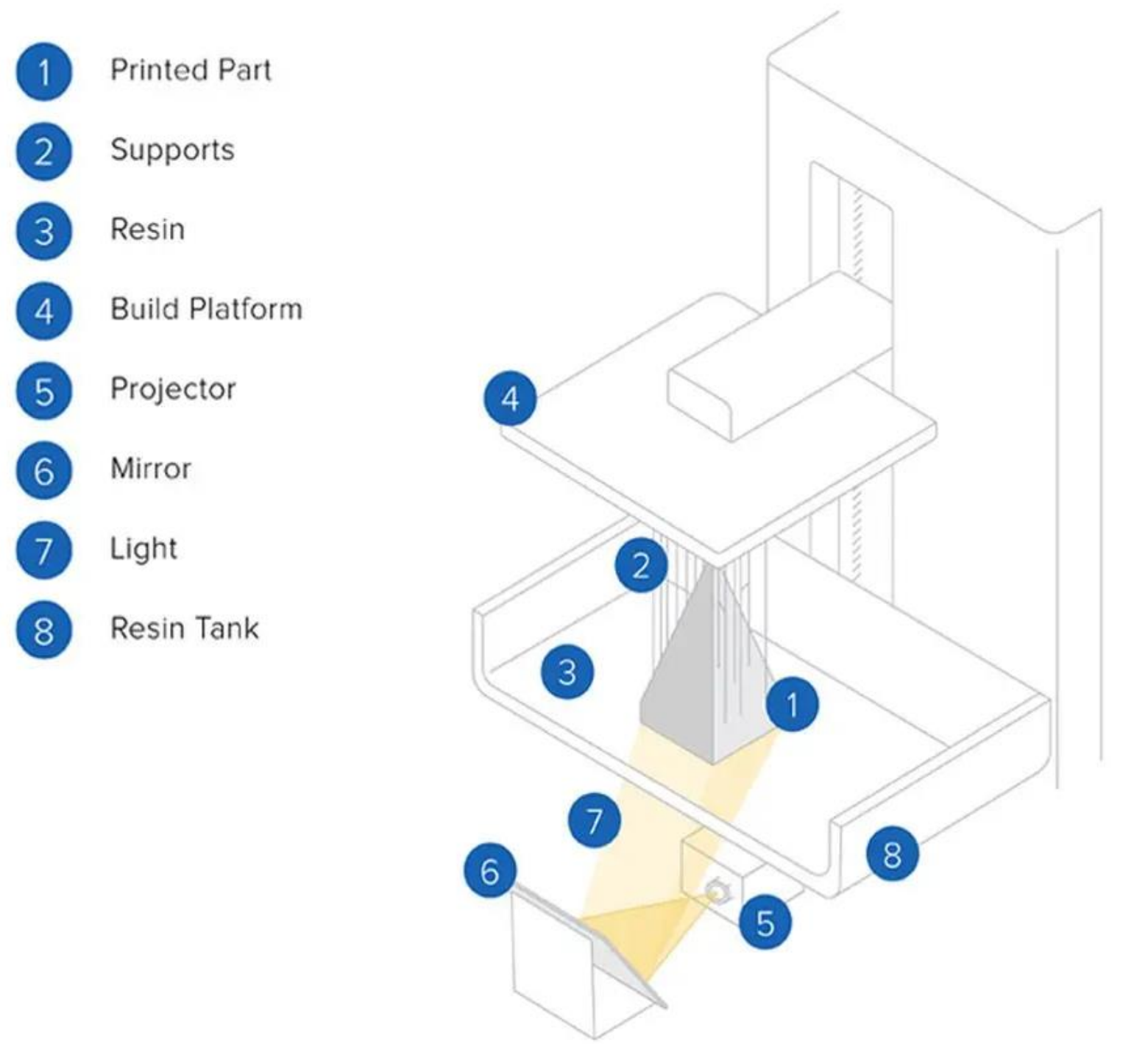


Introduction:

Tribology is the study of wear and friction between interacting surfaces in relative motion. By decreasing the friction between surfaces, parts deteriorate slower and last for longer periods of time. Adding a ceramic coating to metal parts creates a polishing effect that smooths the surfaces and reduces wear. The purpose of this study was to test and characterize the abrasive properties of 3D printed ceramics. The goal was to modify printed parts to have a higher coefficient of friction so that they became more wear resistant. In the study, small testing disks using commercially available 3D ceramic were fabricated using a DLP resin printer. These disks were tested in a Universal Mechanical Tester (UMT) using the Pin-on-Disk method to measure their abrasiveness.

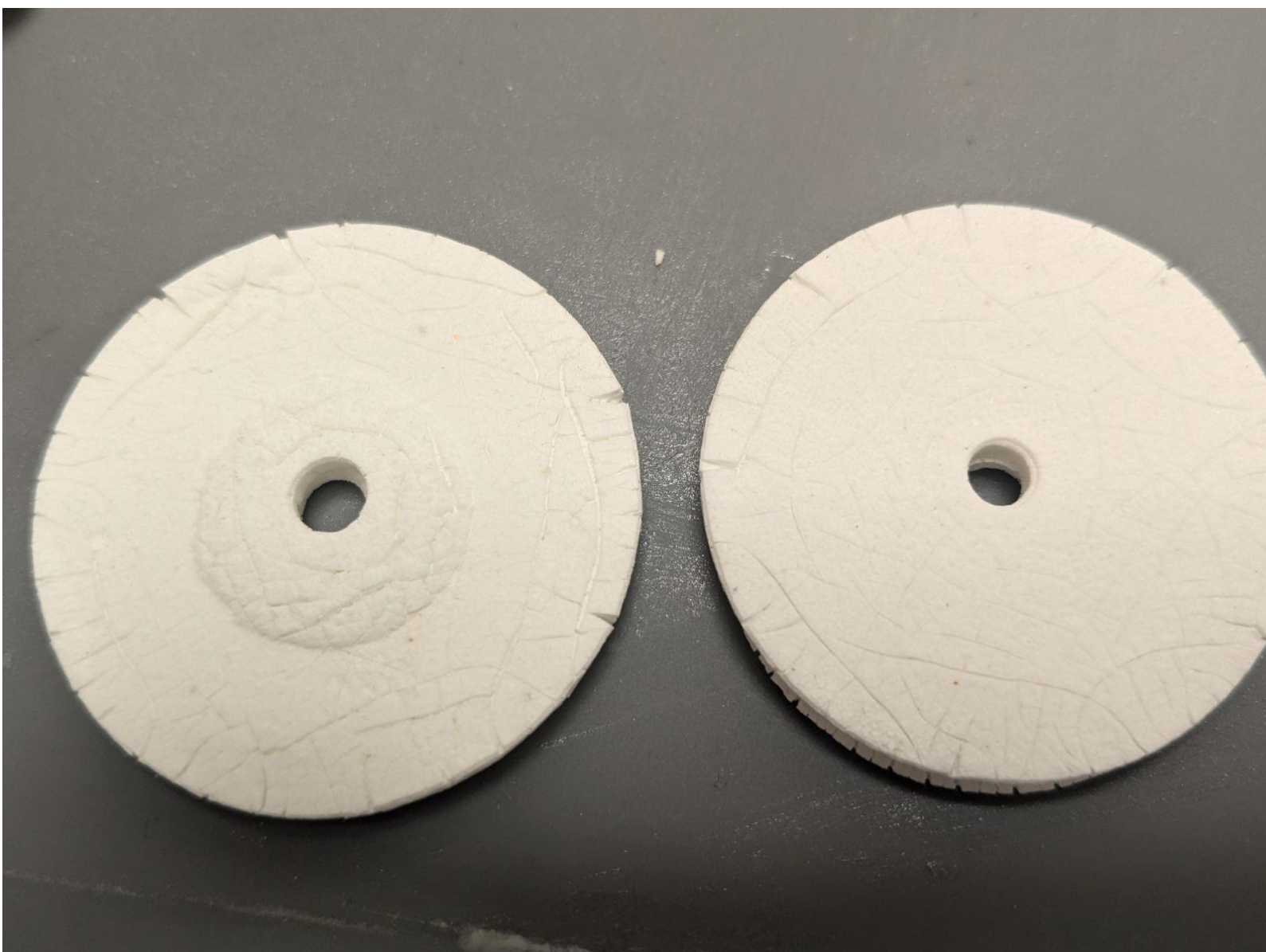
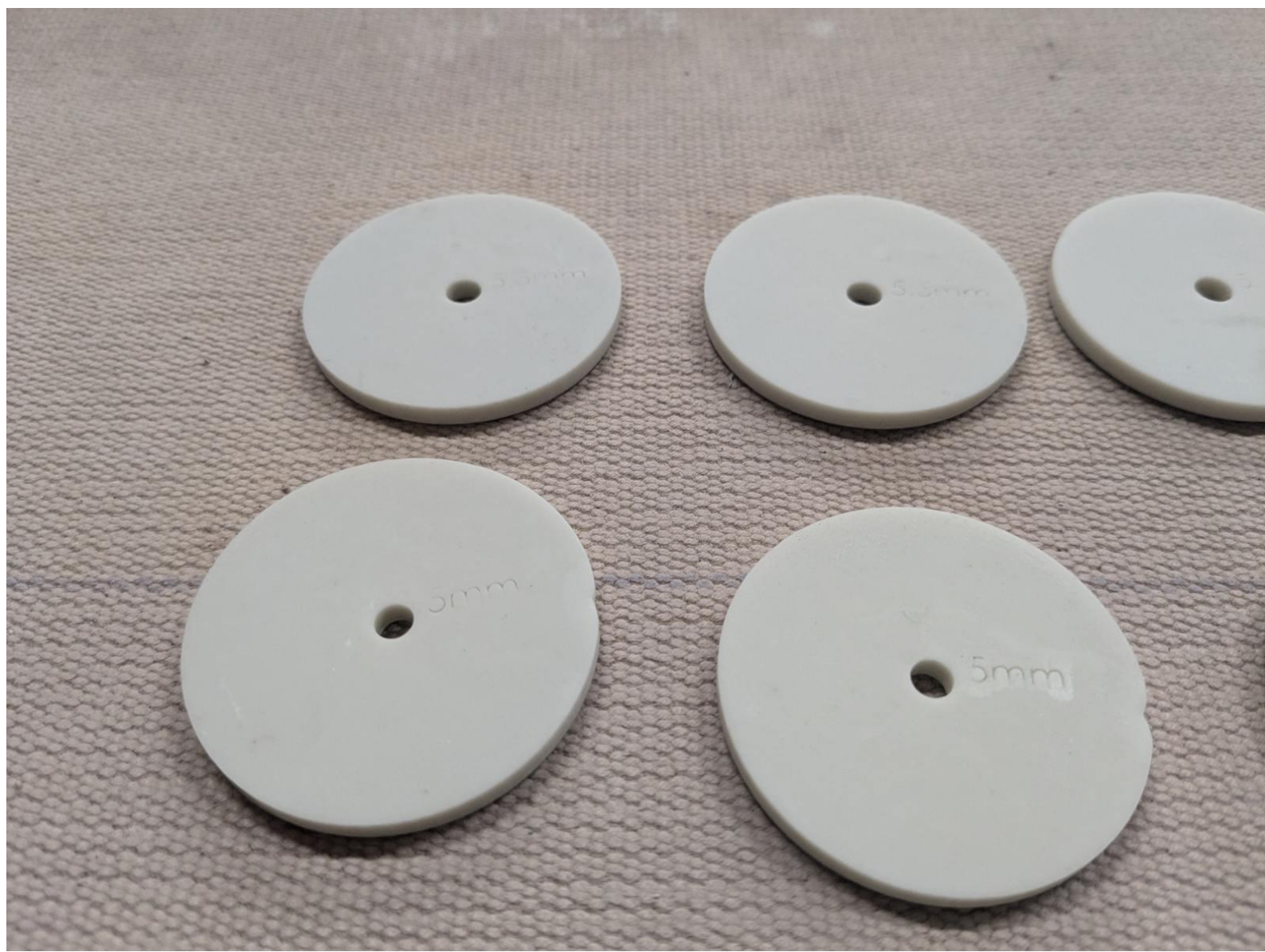
Background:

Digital Light Processing (DLP) is a method of 3D printing that uses ultraviolet light to cure liquid resin into a solid. An object is first broken down into layers. A projector shines an image of each layer into the resin. The object is attached to a build plate, so as each layer cures, it moves up to create a 3D object.



Method:

The disks were printed in a SprintRay MoonRay S100 printer with Tethon 3D Alumina Resin. The disks were printed to be 50mm in diameter with a 5-6.5mm hole in the middle. After printing, the disks were soaked in 99% isopropyl alcohol to remove excess resin and then placed in a UV box to cure any remaining resin. The disks were then placed into a Skutt KM Series Automatic Kiln and fired according to a specific firing sequence.



Testing:

To test the abrasiveness of the disk, a standard pin-on-disk test in the UMT was used. A ¼" ball of 52100 steel was pressed onto the surface of the disk with a force of 1 N. The disk was then rotated so that the ball traveled along the surface of the disk at 50mm/s. The resulting friction between the disk and ball produced a scar on the ball's surface. After a certain number of rotations, the scar on the ball was measured using a microscope. From the size of the scar, the wear volume, or the amount of material lost was calculated.

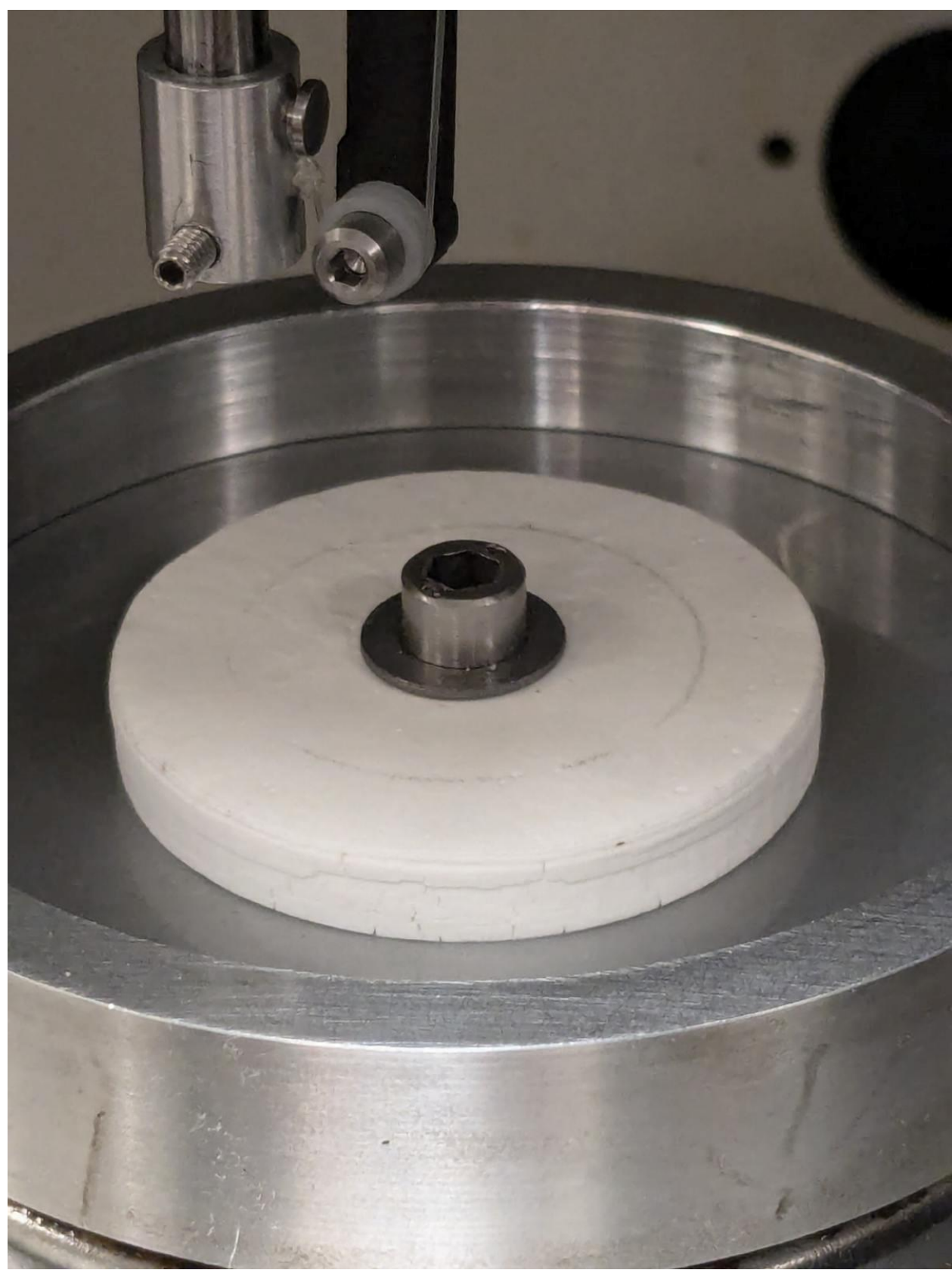
References:

[1] "Formlabs Customer Support," Formlabs Customer Support. Accessed: Sep. 15, 2024. [Online]. Available: <http://support.formlabs.com/>

[2] S. J. Harris and G. G. Krauss, "Improved technique for measuring the ball volume removed in a ball-on-disk test," *Tribology Letters*, vol. 10, no. 3, pp. 187–188, Apr. 2001, doi: [10.1023/A:1009014524815](https://doi.org/10.1023/A:1009014524815).

[3] Z. Chen, J. Li, C. Liu, Y. Liu, J. Zhu, and C. Lao, "Preparation of high solid loading and low viscosity ceramic slurries for photopolymerization-based 3D printing," *Ceramics International*, vol. 45, no. 9, pp. 11549–11557, Jun. 2019, doi: [10.1016/j.ceramint.2019.03.024](https://doi.org/10.1016/j.ceramint.2019.03.024).

[4] "SLA vs. DLP vs. MSLA vs. LCD: Guide to Resin 3D Printers," Formlabs. Accessed: Sep. 15, 2024. [Online]. Available: <https://formlabs.com/blog/resin-3d-printer-comparison-sla-vs-dlp/>



Results:

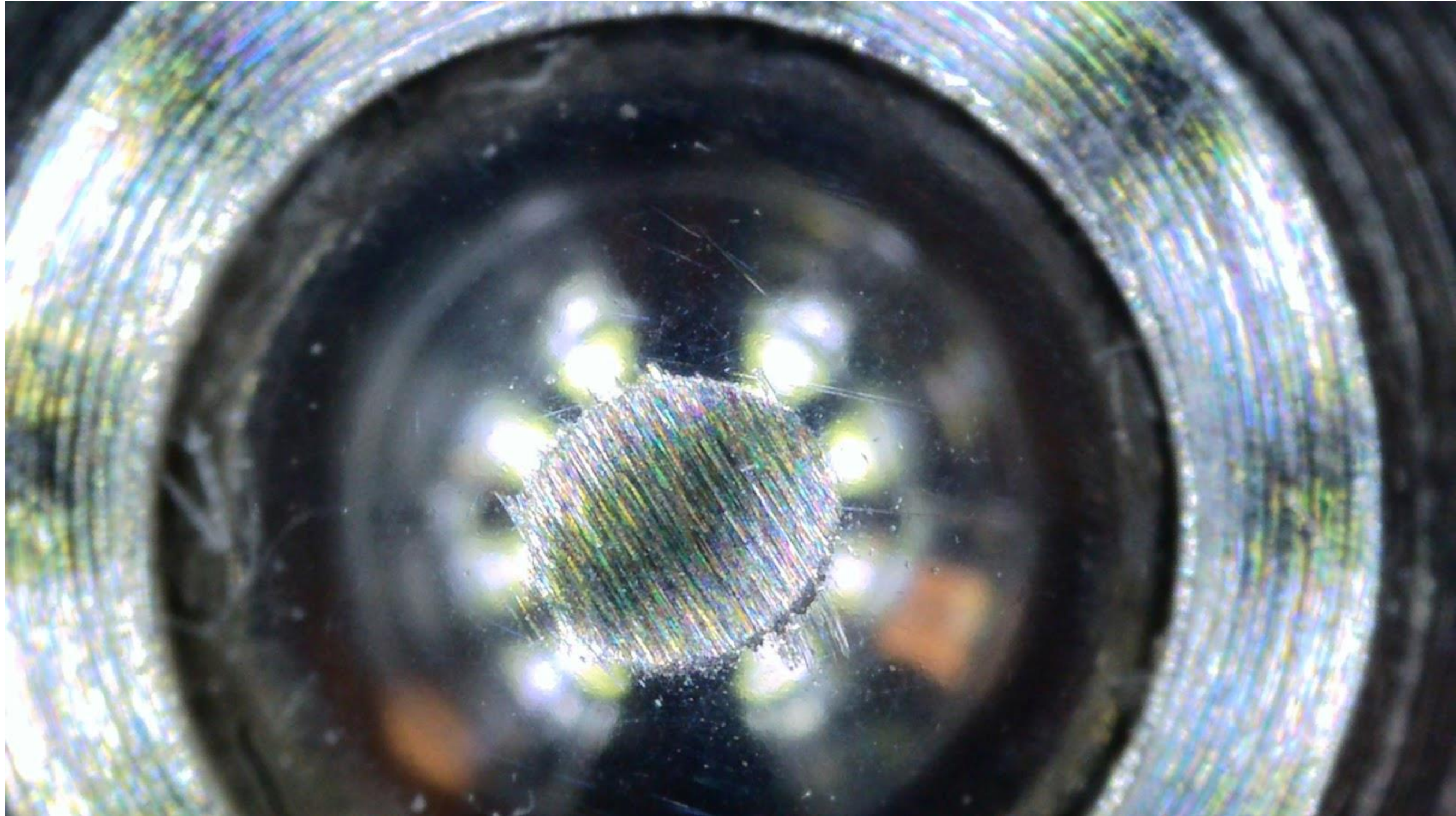
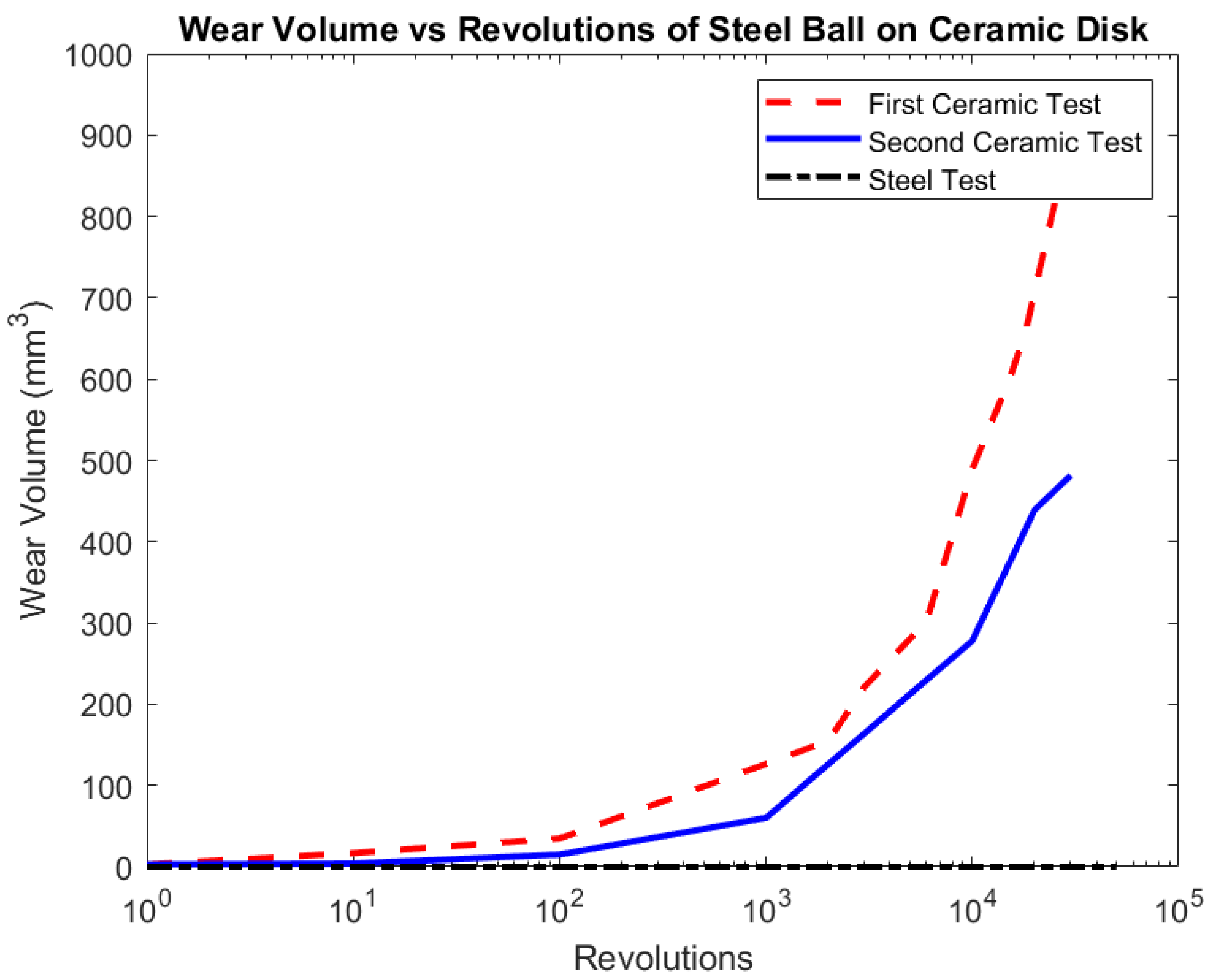
The wear volume of the ball was calculated using the following equation where R is the radius of the ball:

$$V = \pi * \left[h * R^2 - \frac{R^3}{3} + \frac{(R-h)^3}{3} \right]$$

h is the height of the scar given by the Pythagorean theorem,

$$h = R - \sqrt{R^2 - s^2}$$

s is the radius of the scar.



Conclusion:

Based on the results, the ceramic disk is much more abrasive than the steel disk. As the ceramic disk is constantly used, its abrasiveness levels drop. This is most likely due to steel filings becoming embedded in the ceramic and creating a top layer. It is no longer ceramic on steel but more steel-on-steel sliding.

Future Works:

More data needs to be collected, and there is a lot of future research to be done. Future research could include examining how patterns in the disk affect wear and how different added materials, such as metals and low-fire ceramics, impact wear. In addition, research could include finding better ways to measure the scar size on the ball.